



IOT SERVERLESS FLEET MONITORING AND PREDICTIVE MAINTENANCE WITH AMAZON SAGEMAKER

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PROJECT OVERVIEW

- This project focuses on **tracking vehicle health** and performance in real time using AWS serverless services.
- Fleet monitoring is continuously observing vehicles like trucks to **detect issues** such as high temperature, faults, or abnormal behavior.
- Predictive maintenance helps identify problems before they occur, **reducing breakdowns and improving safety**.
- In this system, vehicle temperature raw data is uploaded to **S3**, **Lambda** processes it, **Amazon SageMaker** predicts maintenance needs, **DynamoDB** stores results, and **SNS** sends instant alert emails to the driver/admin when critical issues are detected.

SERVICES USED



AWS Lambda



**Amazon
SageMaker**



Amazon S3



**Amazon
DynamoDB**



**AWS Key
Management
Service**

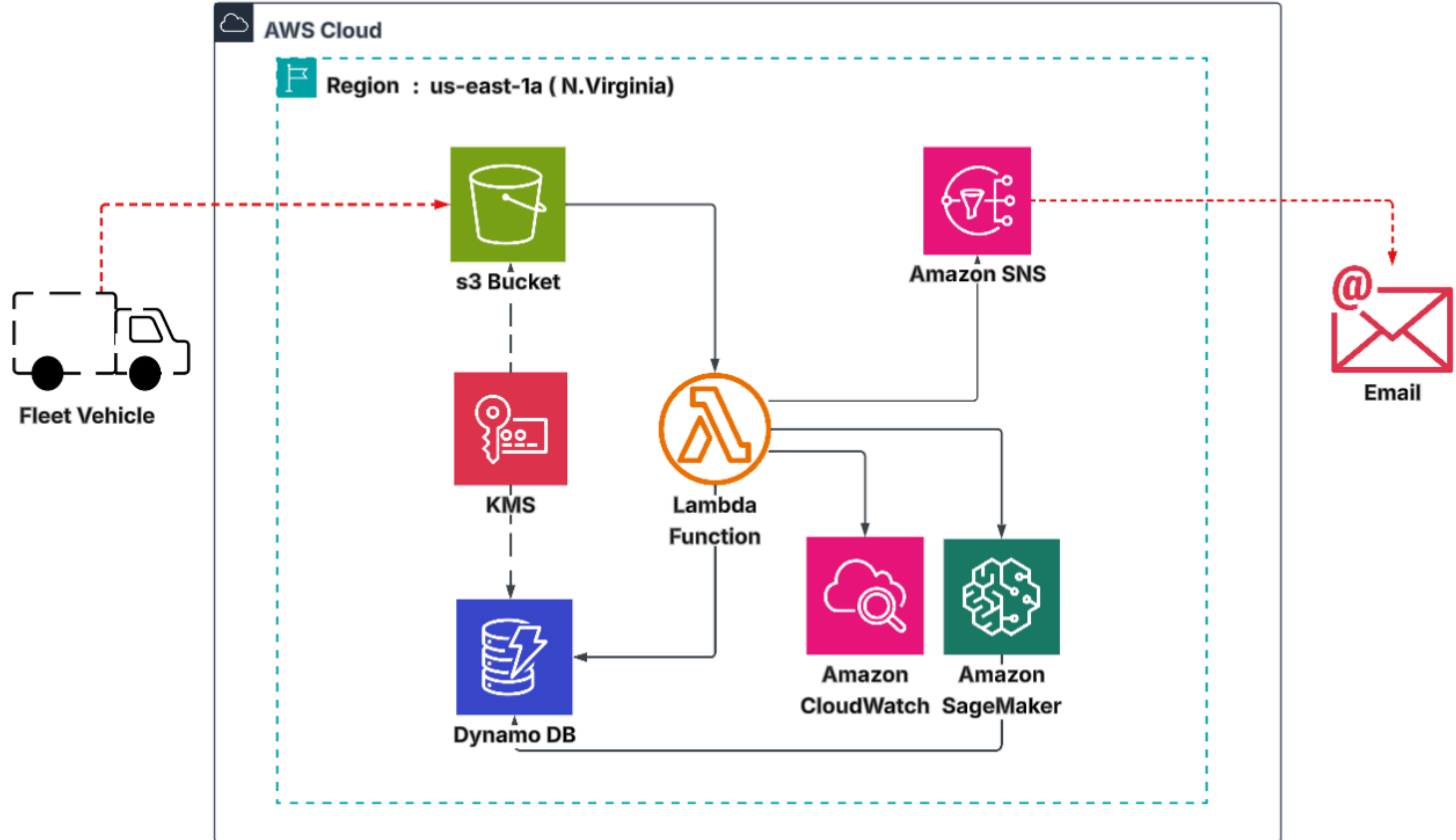


Amazon SNS



**Amazon
CloudWatch**

ARCHITECTURE



IAM ROLE

[fleet-lambda-role](#) > Create policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for a role, attach a policy to it

Search

Allow (7 of 451 services)

Service	Access level	Resource
CloudWatch	Limited: Write	All resources
CloudWatch Logs	Limited: Write	Multiple
DynamoDB	Limited: Read, Write	region string like us-east-1, TableName string like fleet-table
KMS	Limited: Read, Write	region string like us-east-1, KeyID string like fleet-kms
S3	Limited: List, Read, Write	Multiple
SageMaker	Limited: Read	Multiple
SNS	Limited: Write	Multiple

[fleet-sagemaker-role](#) > Create policy

Permissions defined in this policy document specify which actions are allowed or denied. To define permissions for a role, attach a policy to it

Search

Allow (6 of 451 services)

Service	Access level	Resource
CloudWatch	Limited: Write	All resources
CloudWatch Logs	Limited: List, Read, Write	All resources
IAM	Limited: Write	RoleName string like fleet-sagemaker-role
KMS	Limited: Read, Write	region string like us-east-1, KeyID string like fleet-kms
S3	Limited: List, Permissions management, Read, Write, Tagging	Multiple
SageMaker	Full: Tagging Limited: List, Read, Write	All resources

Creating **IAM role** for **Lambda** and **Sagemaker** with permissions for S3,Sagemaker,KMS,Dynamo DB,Cloudwatch and SNS.

S3 BUCKET

VPC

EC2

S3

IAM

Amazon S3 > Buckets > Create bucket

AWS Region

US East (N. Virginia) us-east-1

Bucket type | Info

☒ General purpose
Recommended for most use cases and access patterns. General purpose buckets are the original S3 bucket type. They allow a mix of storage classes that redundantly store objects across multiple Availability Zones.

☐ Directory
Recommended for low-latency use cases. These buckets which provides faster processing of data within a single

Bucket name | Info

fleet-iot-data

Bucket names must be 3 to 63 characters and unique within the global namespace. Bucket names must also begin and end with a letter or number. Valid characters are a-z, 0-9,

Copy settings from existing bucket - optional

Only the bucket settings in the following configuration are copied.

Choose bucket

Format: s3://bucket/prefix

Object Ownership | Info

Control ownership of objects written to this bucket from other AWS accounts and the use of access control lists (ACLs). Object ownership determines who c

Object Ownership

☒ ACLs disabled (recommended)
All objects in this bucket are owned by this account. Access to this bucket and its objects is specified using only policies.

☐ ACLs enabled
Objects in this bucket can be owned by other AWS accounts. Access to this bucket and its objects can be specified using ACLs.

aws

Search

[Alt+S]

VPC

EC2

S3

IAM

Amazon S3

>

Buckets

>

Create bucket

Block Public Access settings for this bucket

Public access is granted to buckets and objects through access control lists (ACLs), bucket policies, access point policies, or all. In order to ensure that public access to this bucket is blocked, turn on Block all public access. These settings apply only to this bucket and its access points. AWS recommends that you turn on Block all public access, but before applying these settings, ensure that your applications will work correctly without public access. If you require some level of public access to this bucket or objects within, you can customize the settings below to suit your specific storage use cases. [Learn more](#)

☒

Block *all* public access

Turning this setting on is the same as turning on all four settings below. Each of the following settings are independent of one another.

☒ **Block public access to buckets and objects granted through *new* access control lists (ACLs)**

S3 will block public access permissions applied to newly added buckets or objects, and prevent the creation of new public access ACLs for existing buckets and objects. This setting doesn't change any existing ACLs that allow public access to S3 resources using ACLs.

☒ **Block public access to buckets and objects granted through *any* access control lists (ACLs)**

S3 will ignore all ACLs that grant public access to buckets and objects.

☒ **Block public access to buckets and objects granted through *new* public bucket or access point policies**

S3 will block new bucket and access point policies that grant public access to buckets and objects. This setting doesn't change any existing policies that allow public access to S3 resources.

☒ **Block public and cross-account access to buckets and objects through *any* public bucket or access point policies**

S3 will ignore public and cross-account access for buckets or access points with policies that grant public access to buckets and objects.

Bucket Versioning

Versioning is a means of keeping multiple variants of an object in the same bucket. You can use versioning to preserve, retrieve, and restore every version of every object stored in the bucket. With versioning, you can easily recover from both unintended user actions and application failures. [Learn more](#)

Bucket Versioning

☐ Disable

☒ Enable

Creating **S3 bucket** with ACLs disabled ,public access blocked and bucket versioning enabled.

KEY MANAGEMENT SERVICE – KMS

aws [Search] [Alt+S] United States (N. Virg)

VPC EC2 S3 IAM

KMS > Customer managed keys > Create key

Step 1 Configure key

Step 2 **Add labels**

Step 3 - optional Define key administrative permissions

Step 4 - optional Define key usage permissions

Step 5 - optional Edit key policy

Step 6 Review

Add labels

You can change the alias at any time. [Learn more](#)

Alias

fleet-kms

Description - optional

You can change the description at any time.

Description

KMS key for fleet monitoring

aws [Search] [Alt+S] United States (N. V

VPC EC2 S3 IAM

KMS > Customer managed keys > Create key

Introducing the new Create key experience
We've improved the create key experience with an enhanced policy editor. [Let us know what you think](#) or you can [use the old experience](#).

Step 1 Configure key

Step 2 Add labels

Step 3 - optional **Define key administrative permissions**

Step 4 - optional Define key usage permissions

Step 5 - optional Edit key policy

Step 6 Review

Define key administrative permissions - optional

Key administrators (1/35)

Select the IAM users and roles authorized to manage this key via the KMS API. These administrators will be added to the key policy. Modifying this Sid might impact the console's ability to update the key policy. [Learn more](#)

Search: priyanka-fl 1 matches

☒	Name	Path	Type
☒	priyanka-fleet	/	User

Key deletion

VPC EC2 S3 IAM

KMS > Customer managed keys

Key Management Service (KMS)

AWS managed keys

Customer managed keys

Custom key stores

AWS CloudHSM key stores

External key stores

Success
Your AWS KMS key was created with alias **fleet-kms** and key ID **9f52ca5b-b444-45c3-b95d-50b9d4929366**. [View key](#)

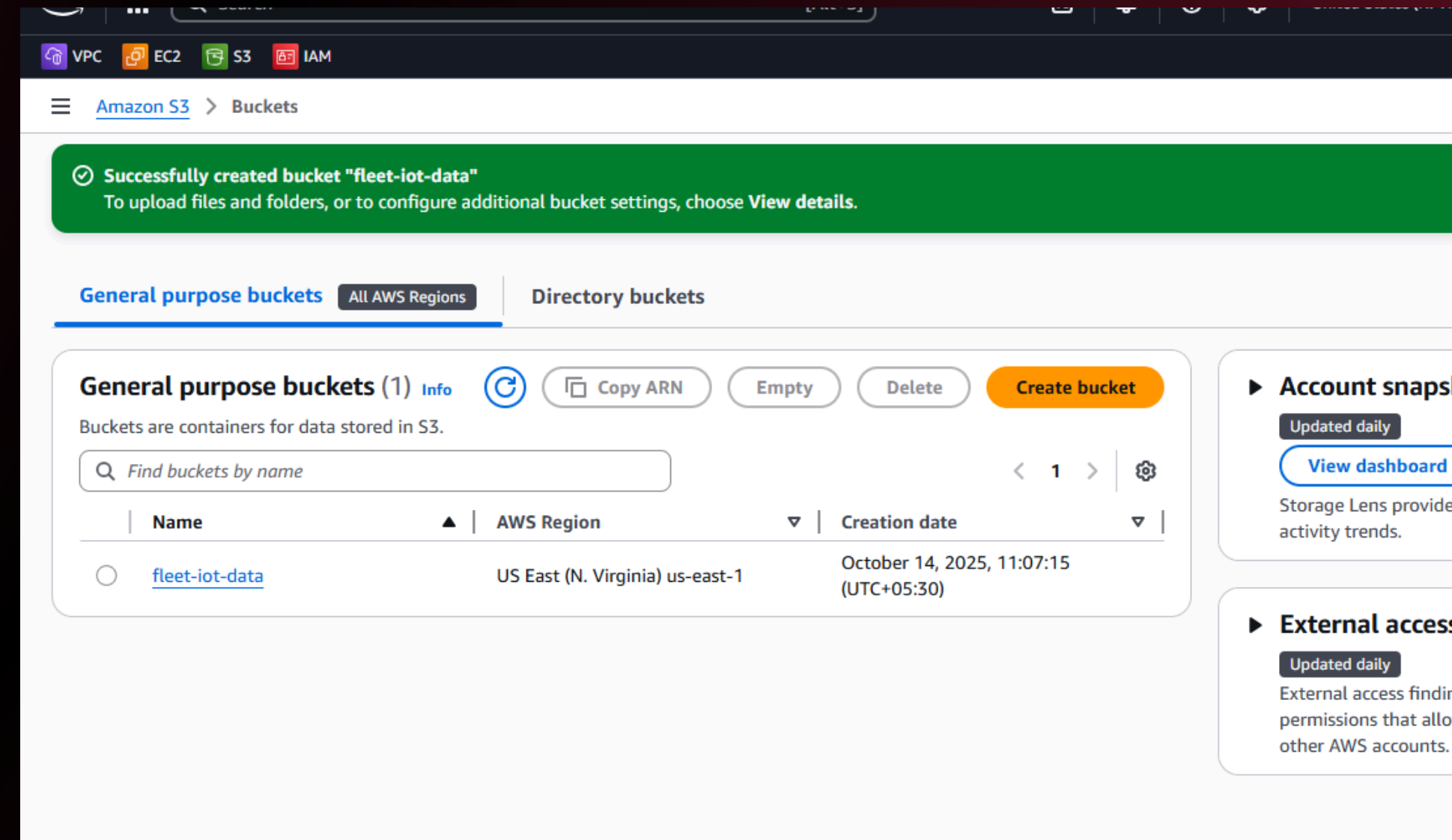
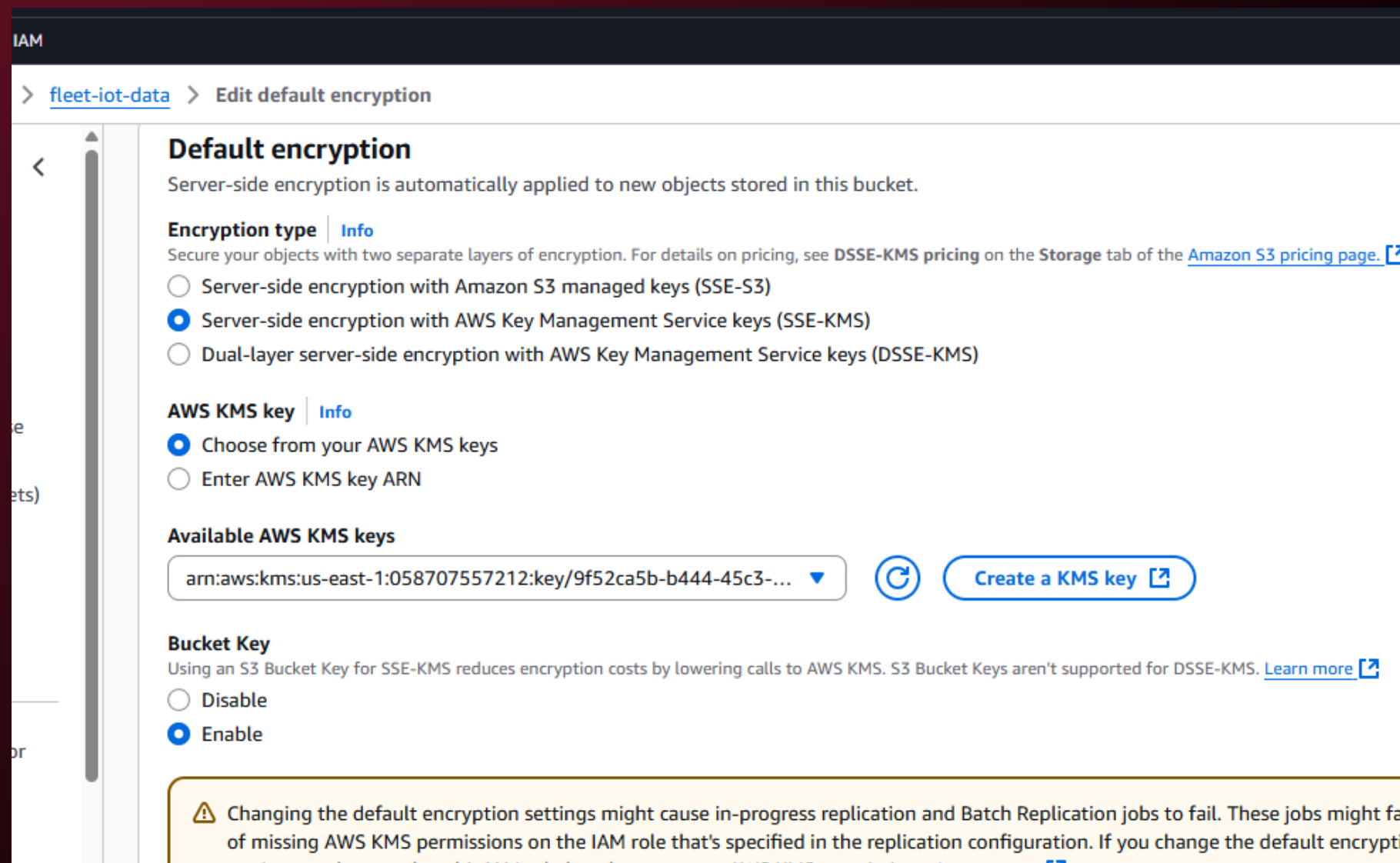
Customer managed keys (1)

Filter keys by properties or tags

☐	Aliases	Key ID	Status	Key type	Key spec
☐	fleet-kms	9f52ca5b-b444-45c3-b9...	Enabled	Symmetric	SYMMETRIC_DEFAU

Creating **KMS key** with permissions given to respective IAM user to **encrypt** and **decrypt S3** and **Dynamo DB**. Permission given to the respective IAM user ID.

ATTACHING KMS KEY TO BUCKET



Attaching the KMS key to S3 bucket created.

DYNAMODB TABLE

Between 3 and 255 characters, containing only letters, numbers, underscores (`_`), hyphens (`-`), and periods (`.`).

Partition key
The partition key is part of the table's primary key. It is a hash value that is used to retrieve items from your table and allocate data across hosts for scalability and availability.

`device_id` String

1 to 255 characters and case sensitive.

Sort key - optional
You can use a sort key as the second part of a table's primary key. The sort key allows you to sort or search among all items sharing the same partition key.

`timestamp` String

1 to 255 characters and case sensitive.

Table settings

☒ **Default settings**
The fastest way to create your table. You can modify most of these settings after your table has been created. To modify these settings now, choose 'Customize settings'.

☐ **Customize settings**
Use these advanced features to make DynamoDB...

All user data stored in Amazon DynamoDB is fully encrypted at rest. By default, Amazon DynamoDB manages the encryption key, and you are not charged for this service.

Encryption key management

☐ **AWS owned key** [Learn more](#)
The key is owned and managed by DynamoDB. You are not charged additional fees for using this key.

☐ **AWS managed key** [Learn more](#)
The key is stored in your account and managed by AWS Key Management Service (KMS). AWS KMS charges apply.

☒ **Customer managed key** [Learn more](#)
The key is stored in your account and managed by you. AWS KMS charges apply.

`arn:aws:kms:us-east-1:058707557212:key/9f52ca5b-b444-45c3-b95d-5...`

Deletion protection [Info](#)

☒ Deletion protection is turned off by default. Deletion protection protects the table from being deleted unintentionally. You can turn on deletion protection after the table has been created.

☐ Turn on deletion protection

DynamoDB

Dashboard
Tables
Explore items
PartiQL editor
Backups
Exports to S3
Imports from S3

Tables (1) [Info](#)

☐ **Name** ☐ **Status** ☐ **Partition key** ☐ **Sort key** ☐ **Indexes** ☐ **Replication Regions** ☐ **Deletion protection** ☐ **Favorite**

☐	fleet-table	✓ Active	device_id (S)	timestamp (S)	0	0	Off	☆
--------------------------	-----------------------------	----------	---------------	---------------	---	---	-----	---

Created **Dynamo DB table** with the required **partition key** and **Source key** ,added the **KMS key** as well for encryption and decryption.

AMAZON SNS

The screenshot shows the 'Create topic' page in the AWS Management Console. The breadcrumb navigation is 'Amazon SNS > Topics > Create topic'. The page title is 'Create topic'. Under the 'Details' section, the 'Type' is set to 'Standard' (selected with a radio button). The 'FIFO (first-in, first-out)' option is also visible but not selected. The 'Name' field contains 'fleet-sns'. The 'Display name - optional' field contains 'fleetmonitor-sns'. The 'Encryption - optional' section is expanded, showing that Amazon SNS provides in-transit encryption by default and enabling server-side encryption adds at-rest encryption to the topic.

Details

Type | [Info](#)
Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ Standard

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

fleet-sns

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - optional | [Info](#)
To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

fleetmonitor-sns

Maximum 100 characters.

► **Encryption - optional**

Amazon SNS provides in-transit encryption by default. Enabling server-side encryption adds at-rest encryption to your topic.

The screenshot shows the 'Create subscription' page in the AWS Management Console. The breadcrumb navigation is 'Amazon SNS > Subscriptions > Create subscription'. The page title is 'Create subscription'. Under the 'Details' section, the 'Topic ARN' field contains 'arn:aws:sns:us-east-1:058707557212:fleet-sns'. The 'Protocol' is set to 'Email'. The 'Endpoint' field contains 'priyankaraj0919@gmail.com'. A note states: 'After your subscription is created, you must confirm it. Info'. The 'Subscription filter policy - optional' section is expanded, showing that this policy filters the messages that a subscriber receives.

Details

Topic ARN

arn:aws:sns:us-east-1:058707557212:fleet-sns

Protocol
The type of endpoint to subscribe

Email

Endpoint
An email address that can receive notifications from Amazon SNS.

priyankaraj0919@gmail.com

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

Created **SNS topic**, added **mail address** under **subscription** to receive fleet alert mail.

AMAZON SNS – TOPICS AND SUBSCRIPTIONS

The image displays two screenshots of the Amazon SNS console interface. The top screenshot shows the 'Topics' page, and the bottom screenshot shows the 'Subscriptions' page. Both pages include a left-hand navigation menu, a top header with account information, and a main content area with a table of items.

Amazon SNS Topics (3)

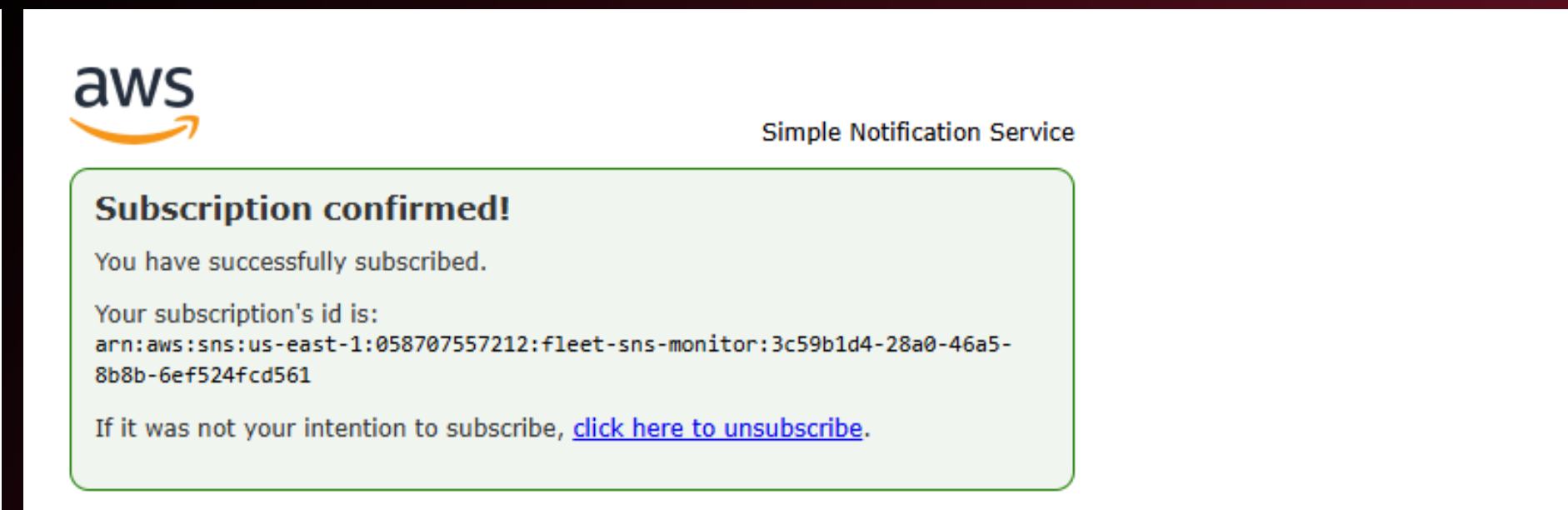
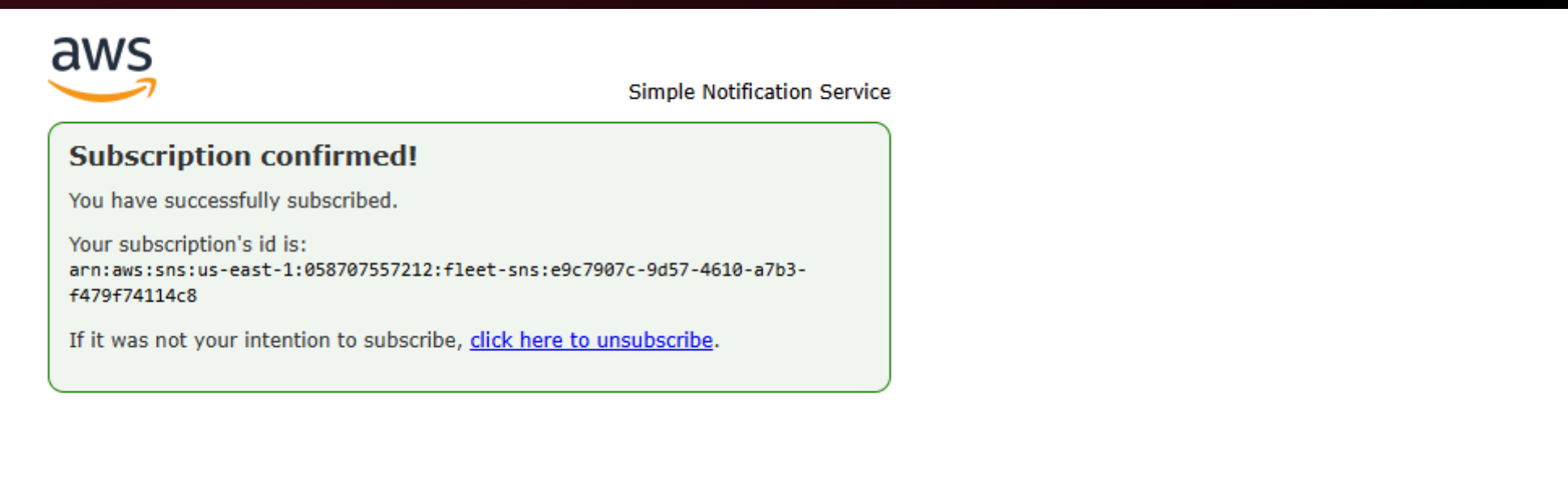
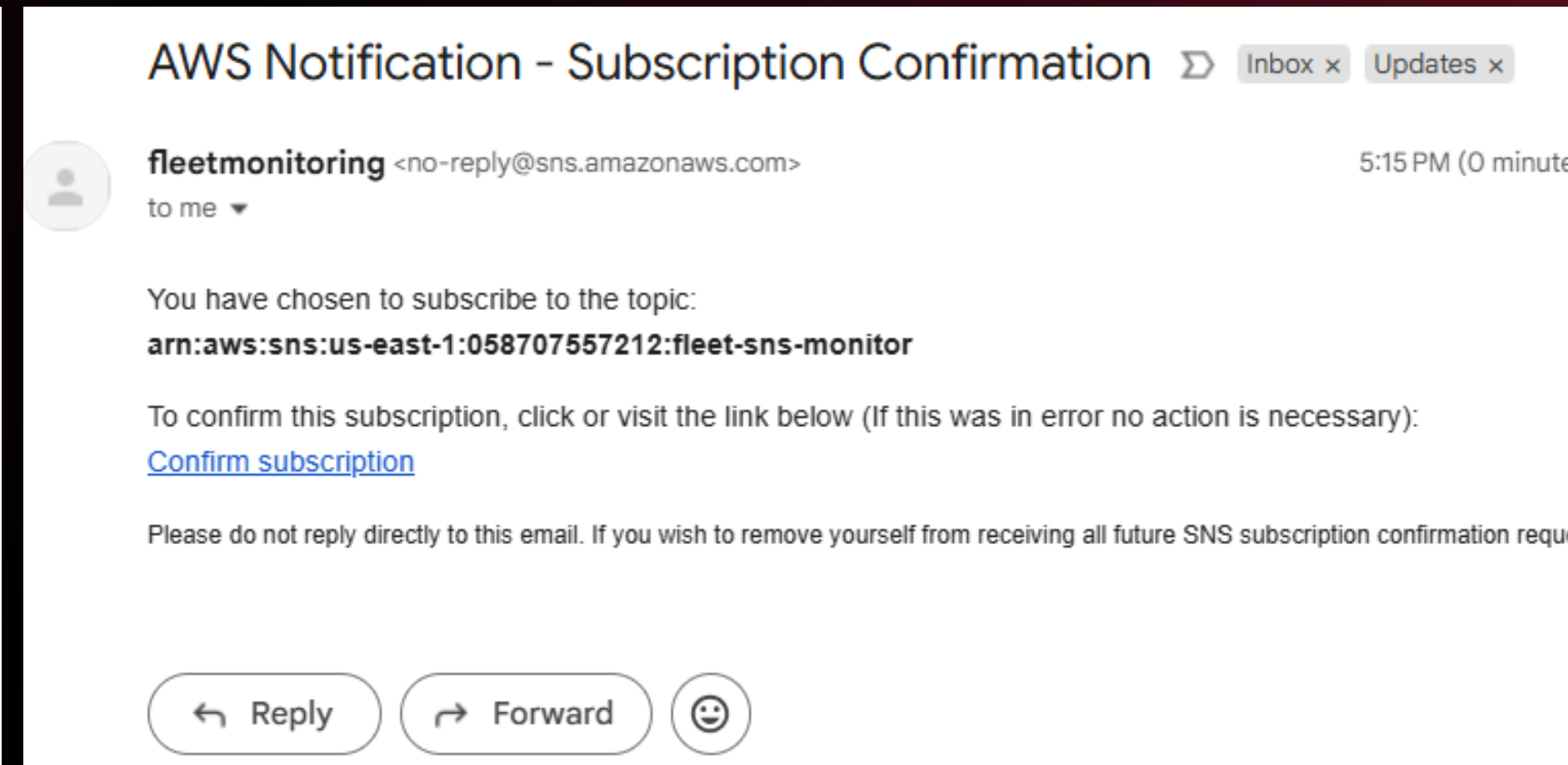
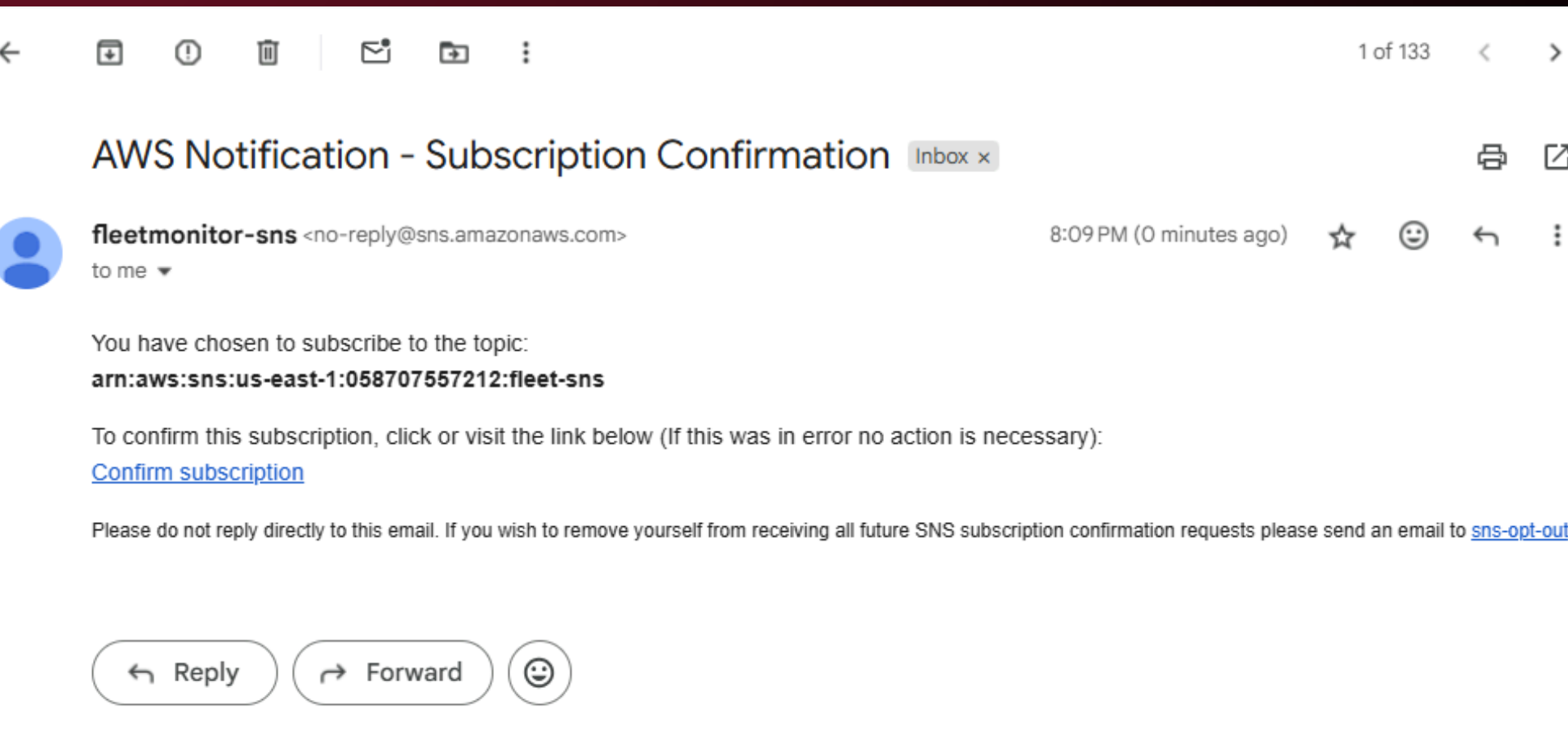
Name	Type	ARN
Billing-alarm	Standard	arn:aws:sns:us-east-1:058707557212:Billin...
fleet-sns	Standard	arn:aws:sns:us-east-1:058707557212:fleet-...
fleet-sns-monitor	Standard	arn:aws:sns:us-east-1:058707557212:fleet-...

Amazon SNS Subscriptions (2)

ID	Endpoint	Status	Protocol	Topic
3c59b1d4-28a0-46a5-...	[redacted]2001@gm...	Confirmed	EMAIL	fleet-sns-monitor
e9c7907c-9d57-4610-...	priyankaraj0919@gma...	Confirmed	EMAIL	fleet-sns

Added 2 topics and 2 Subscription mail address to check the demo.

CONFIRMATION MAIL



An **Automated email** for subscription is received, **confirmed subscription** to get the alert mails.

SAGEMAKER ENDPOINT

The screenshot shows the AWS SageMaker Studio interface for deploying a model to an endpoint. The breadcrumb navigation at the top reads: SageMaker Studio > Inference Experience > Models > Deploy. The left sidebar contains a menu with 'Applications (5)' (JupyterLab, RStudio, Canvas, Code Editor, MLflow), 'Partner AI Apps' (marked as 'New'), and navigation links for Home, Running instances, Compute, Data, Auto ML, Experiments, and Jobs. The main content area is titled 'Deploy model to endpoint' with a subtitle 'Deploy your models to a SageMaker endpoint by selecting the deployment resources. Learn more'. Below this is the 'Endpoint settings' section, which includes: 'Endpoint name' (a text input field with placeholder 'Enter endpoint name'), 'Custom endpoint name' (a text input field containing 'fleet-alert-ai'), 'Instance type' (a dropdown menu set to 'ml.g5.xlarge' with a note 'Default Instance type is changed'), and 'Initial instance count' (a text input field set to '1'). The 'Inference type' is set to 'Real-time' with a description: 'For sustained traffic and consistently low latency. Supports payload sizes up to 6 MB and runtimes up to 60 sec.' Below these settings is a 'Models' section. At the bottom right of the main area are 'Cancel' and 'Deploy' buttons. The footer contains links for Privacy, Site Terms, and Cookie Preferences, along with the copyright notice: '© 2025, Amazon Web Services, Inc. or its affiliates. All rights reserved.'

Creating **SageMaker endpoint** to be invoked by **Lambda** using a simple ML classification / regression model for **anomaly detection on temperature data**

AWS LAMBDA

The screenshot shows the 'Create function' page in the AWS Lambda console. The breadcrumb navigation is 'Lambda > Functions > Create function'. A text input field contains 'fleet-lambda'. Below it, a message states: 'Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).' The 'Runtime' section is set to 'Python 3.13'. The 'Architecture' section has 'x86_64' selected. The 'Permissions' section shows a default role. At the bottom, the 'Change default execution role' section has 'Use an existing role' selected, and a dropdown menu shows 'fleet-lambda-role'.

Enter a name that describes the purpose of your function.

fleet-lambda

Function name must be 1 to 64 characters, must be unique to the Region, and can't include spaces. Valid characters are a-z, A-Z, 0-9, hyphens (-), and underscores (_).

Runtime | Info
Choose the language to use to write your function. Note that the console code editor supports only Node.js, Python, and Ruby.

Python 3.13

Architecture | Info
Choose the instruction set architecture you want for your function code.

☐ arm64
☒ x86_64

Permissions | Info
By default, Lambda will create an execution role with permissions to upload logs to Amazon CloudWatch Logs. You can customize this default role later when adding triggers.

▼ **Change default execution role**

Execution role
Choose a role that defines the permissions of your function. To create a custom role, go to the [IAM console](#).

☐ Create a new role with basic Lambda permissions
☒ Use an existing role
☐ Create a new role from AWS policy templates

Existing role
Choose an existing role that you've created to be used with this Lambda function. The role must have permission to upload logs to Amazon CloudWatch Logs.

fleet-lambda-role

The screenshot shows the 'fleet-lambda' page in the AWS Lambda console. A green banner at the top says: 'Successfully created the function fleet-lambda. You can now change its code and configuration. To invoke your function with a test event, choose "Test".' The breadcrumb navigation is 'Lambda > Functions > fleet-lambda'. The 'Function overview' section shows the function name 'fleet-lambda' and 'Layers (0)'. There are buttons for 'Add trigger' and 'Add destination'. The right sidebar shows 'Description', 'Last modified 14 seconds ago', 'Function ARN', and 'Function URL'. The bottom navigation bar includes 'Code', 'Test', 'Monitor', 'Configuration', 'Aliases', and 'Versions'.

Successfully created the function **fleet-lambda**. You can now change its code and configuration. To invoke your function with a test event, choose "Test".

fleet-lambda

▼ **Function overview** | Info

Diagram | Template

fleet-lambda

Layers (0)

+ Add trigger

+ Add destination

Description
-

Last modified
14 seconds ago

Function ARN
[arn:aws:lambda:us-east-1:123456789012:function:fleet-lambda](#)

Function URL | Info
-

Code | Test | Monitor | Configuration | Aliases | Versions

Code source | Info

[Open in Visual Studio Code](#)

Created **Lambda function** with Runtime **Python 3.13**
and attached the respective role created.

S3 TRIGGER FOR LAMBDA

aws Search [Alt+S] VPC EC2 S3 IAM

Lambda > Add triggers

Add trigger

Trigger configuration Info

 **S3**
aws asynchronous storage

Bucket
Choose or enter the ARN of an S3 bucket that serves as the event source. The bucket must be in the same region as the function.

Bucket region: us-east-1

Event types
Select the events that you want to have trigger the Lambda function. You can optionally set up a prefix or suffix for an event. However, for each bucket, individual events cannot have prefixes or suffixes that could match the same object key.

☒ All object create events 

Prefix - optional
Enter a single optional prefix to limit the notifications to objects with keys that start with matching characters. Any [special characters](#) must be URL encoded.

Suffix - optional
Enter a single optional suffix to limit the notifications to objects with keys that end with matching characters. Any [special characters](#) must be URL encoded.

aws Search [Alt+S] VPC EC2 S3 IAM Lambda Amazon SageMaker AI Simple Notification Service

Lambda > Functions > fleet-lambda

Code Test Monitor **Configuration** Aliases Versions

General configuration

Triggers

Permissions

Destinations

Function URL

Environment variables

Tags

VPC

RDS databases

Monitoring and operations tools

Concurrency and recursion detection

Asynchronous invocation

Triggers (1) Info    

☐ | **Trigger**

 **S3: [fleet-iot-data](#)**
arn:aws:s3:::fleet-iot-data

▼ Details

Bucket arn: **arn:aws:s3:::fleet-iot-data**
Event types: **s3:ObjectCreated:***
isComplexStatement: **No**
Notification name: **eebe395e-bfb0-4451-a696-7f7630afb5c4**
Service principal: **s3.amazonaws.com**
Source account: **058707557212**
Statement ID: **lambda-e31f55a5-c005-410a-9dab-befade9c282c, 058707557212_event_permissions_from_fleet-iot-data-lambda, AllowS3Invoke**

Added **S3 trigger** to lambda, with the required event and permissions.

LAMBDA FUNCTION

```
VPC EC2 S3 IAM Lambda Amazon SageMaker AI Simple Notification Service

Lambda > Functions > fleet-lambda

lambda_function.py
1 import boto3
2 import json
3 import re
4 from decimal import Decimal
5
6 s3_client = boto3.client('s3', region_name='us-east-1')
7 dynamodb = boto3.resource('dynamodb', region_name='us-east-1')
8 sns_client = boto3.client('sns', region_name='us-east-1')
9 cloudwatch = boto3.client('cloudwatch', region_name='us-east-1')
10 sagemaker_runtime = boto3.client('sagemaker-runtime', region_name='us-east-1')
11
12 DYNAMO_TABLE_NAME = 'fleet-table'
13 SAGEMAKER_ENDPOINT = 'fleet-alert-ai'
14 FLEET_BUCKET = 'fleet-iot-data'
15 FLEET_FILE = 'fleet_data.txt'
16
17 VEHICLE_TO_SNS = {
18     'VHC001': 'arn:aws:sns:us-east-1:058707557212:fleet-sns',
19     'VHC002': 'arn:aws:sns:us-east-1:058707557212:fleet-sns-monitor'
20 }
21
22 table = dynamodb.Table(DYNAMO_TABLE_NAME)
23
24 def convert_floats_to_decimal(obj):
25     if isinstance(obj, float):
26         return Decimal(str(obj))
27     elif isinstance(obj, dict):
28         return {k: convert_floats_to_decimal(v) for k, v in obj.items()}
```

Create new test event

Event Name
Testevent1
Maximum of 25 characters consisting of letters, numbers, dots, hyphens and underscores.

Event sharing settings
☒ Private
This event is only available in the Lambda Console and to the event creator. You can configure a total of ten. [Learn more](#)
☐ Shareable
This event is available to IAM users within the same account who have permissions to access and use shareable events. [Learn more](#)

Template - optional
Testevent1

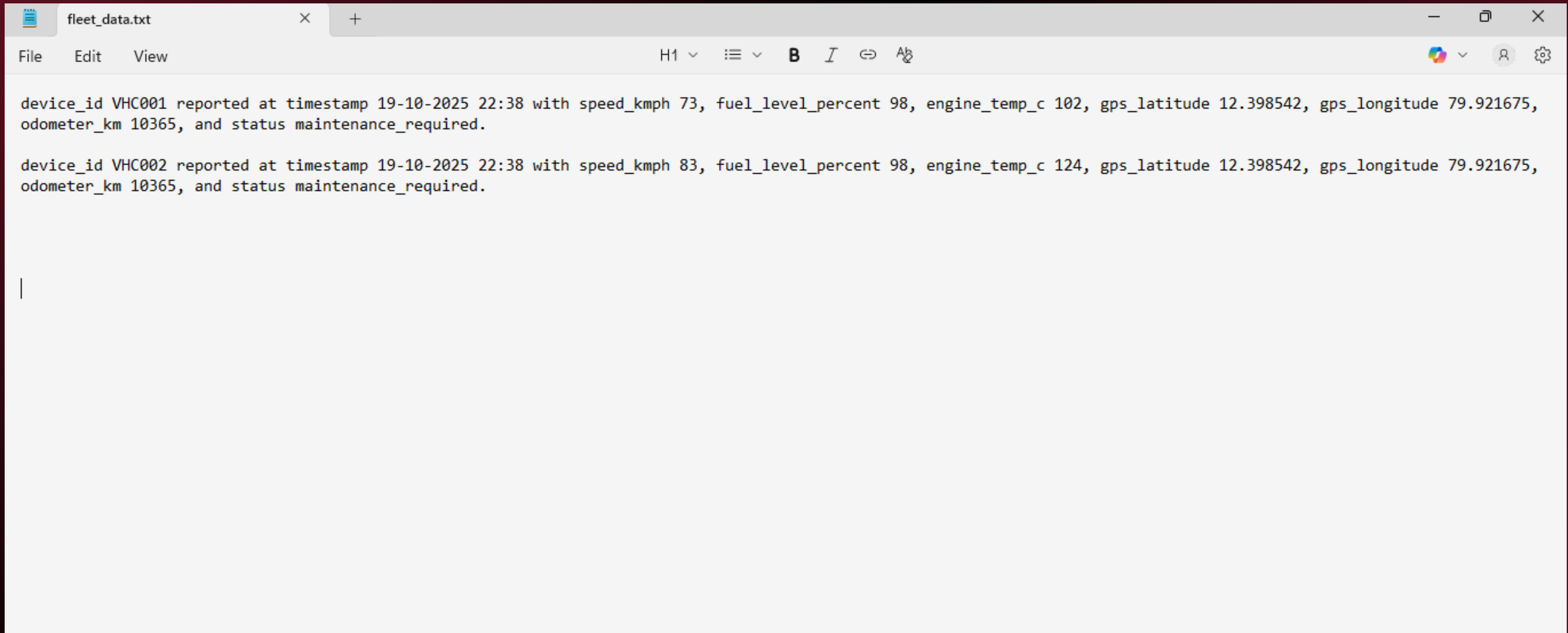
Event JSON

```
1 {
2   "Records": [
3     {
4       "s3": {
5         "bucket": {
6           "name": "fleet-iot-data"
7         },
8       "object": {
```

Test event is saved successfully.

Lambda orchestrates the **complete workflow** using python ,ingesting data from **S3**, sending alerts, invoking SageMaker for ML **predictions**, and storing results in **DynamoDB** .For full code - [Github](#)

RAW DATA



The image shows a text editor window titled 'fleet_data.txt'. The editor has a menu bar with 'File', 'Edit', and 'View'. The toolbar includes a font size dropdown set to 'H1', a list icon, bold, italic, link, and unlink buttons. The text area contains two lines of raw data, each representing a vehicle report. The first line is for device_id VHC001, and the second is for device_id VHC002. Both reports include a timestamp of 19-10-2025 22:38, speed_kmph, fuel_level_percent, engine_temp_c, gps_latitude, gps_longitude, odometer_km, and status maintenance_required.

```
device_id VHC001 reported at timestamp 19-10-2025 22:38 with speed_kmph 73, fuel_level_percent 98, engine_temp_c 102, gps_latitude 12.398542, gps_longitude 79.921675, odometer_km 10365, and status maintenance_required.  
  
device_id VHC002 reported at timestamp 19-10-2025 22:38 with speed_kmph 83, fuel_level_percent 98, engine_temp_c 124, gps_latitude 12.398542, gps_longitude 79.921675, odometer_km 10365, and status maintenance_required.  
  
|
```

Using 2 raw sentence datas for demo.

WORKING DEMO

The screenshot shows the AWS S3 console's 'Upload' page. The breadcrumb trail is 'Amazon S3 > Buckets > fleet-iot-data > Upload'. The page title is 'Upload' with an 'Info' link. A message states: 'Add the files and folders you want to upload to S3. To upload a file larger than 160GB, use the AWS CLI, AWS SDKs or Amazon S3 REST API. [Learn more](#)'. Below this is a dashed box with the text: 'Drag and drop files and folders you want to upload here, or choose **Add files** or **Add folder**.' The 'Files and folders' section shows '1 total, 450.0 B'. A search bar contains 'Find by name'. A table lists the upload items:

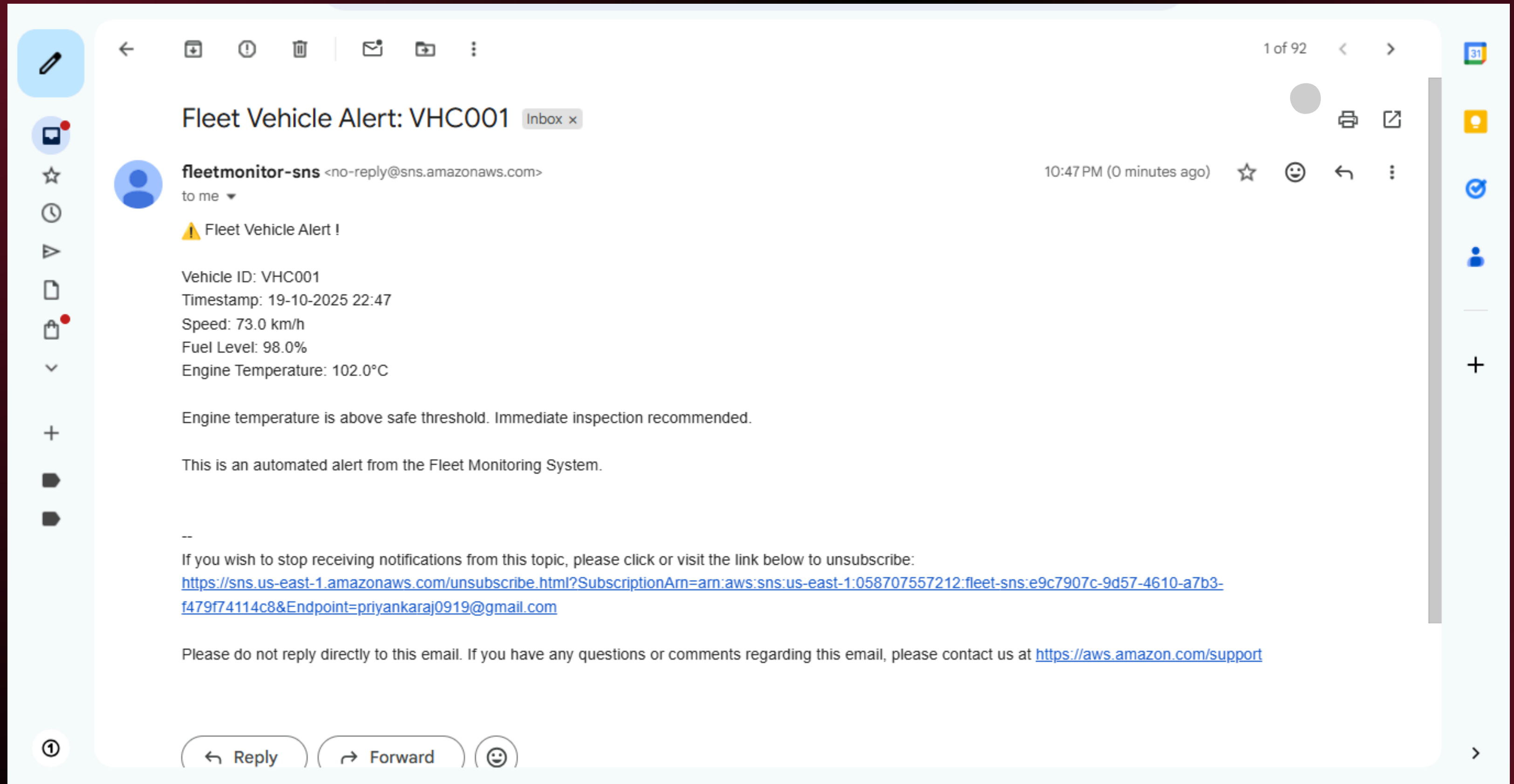
☐	Name	Folder	Type	Size
☐	fleet_data.txt	-	text/plain	450.0 B

Buttons for 'Remove', 'Add files', and 'Add folder' are at the top right of the table. The 'Destination' section shows 'Destination: s3://fleet-iot-data'. Below the upload interface, a green success message states: 'Upload succeeded. For more information, see the **Files and folders** table.' The 'Upload: status' section has a 'Close' button and a message: 'After you navigate away from this page, the following information is no longer available.' The 'Summary' section shows:

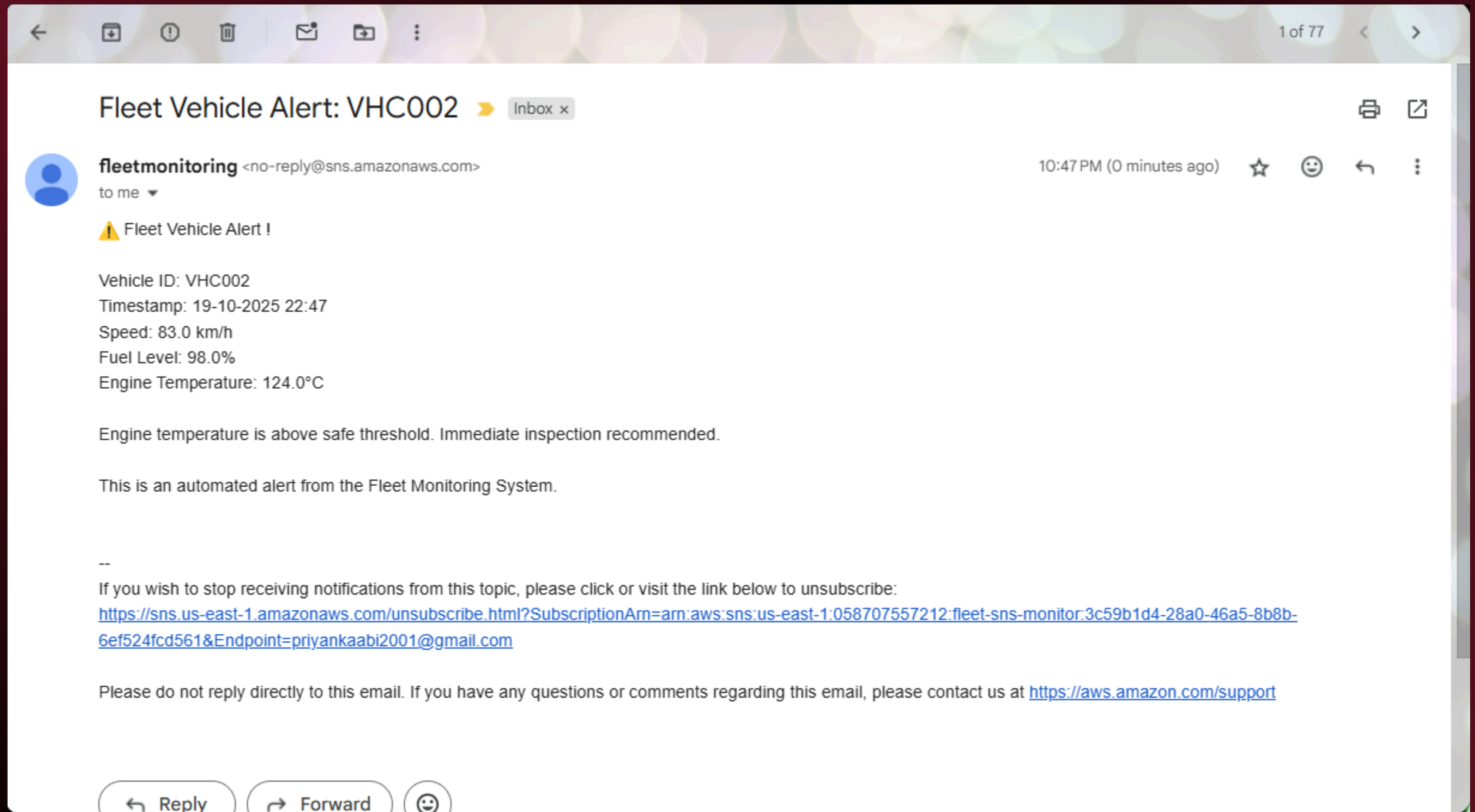
Destination	Succeeded	Failed
s3://fleet-iot-data	1 file, 450.0 B (100.00%)	0 files, 0 B (0%)

Uploading the raw data file to s3 bucket.

FLEET ALERT MAIL



Received the **alert mail** to the **1st Subscription mail** id as the temperature is **>100 degree**.



Received the **alert mail** to the **2st Subscription mail** id added as the temperature is **>100 degree**.

SAGEMAKER PREDICTION

The screenshot displays the AWS Management Console interface for the SageMaker prediction task. The top navigation bar shows the AWS logo, a search bar, and various service icons including VPC, EC2, S3, IAM, Lambda, Amazon SageMaker AI, and Simple Notification Service. The user is logged in as 'priyanka-fleet' in the 'United States (N. Virginia)' region.

The main content area is titled 'DynamoDB > Explore items > fleet-table'. It features two dropdown menus: 'Select a table or index' (set to 'Table - fleet-table') and 'Select attribute projection' (set to 'All attributes'). Below these are 'Filters - optional' and buttons for 'Run' and 'Reset'.

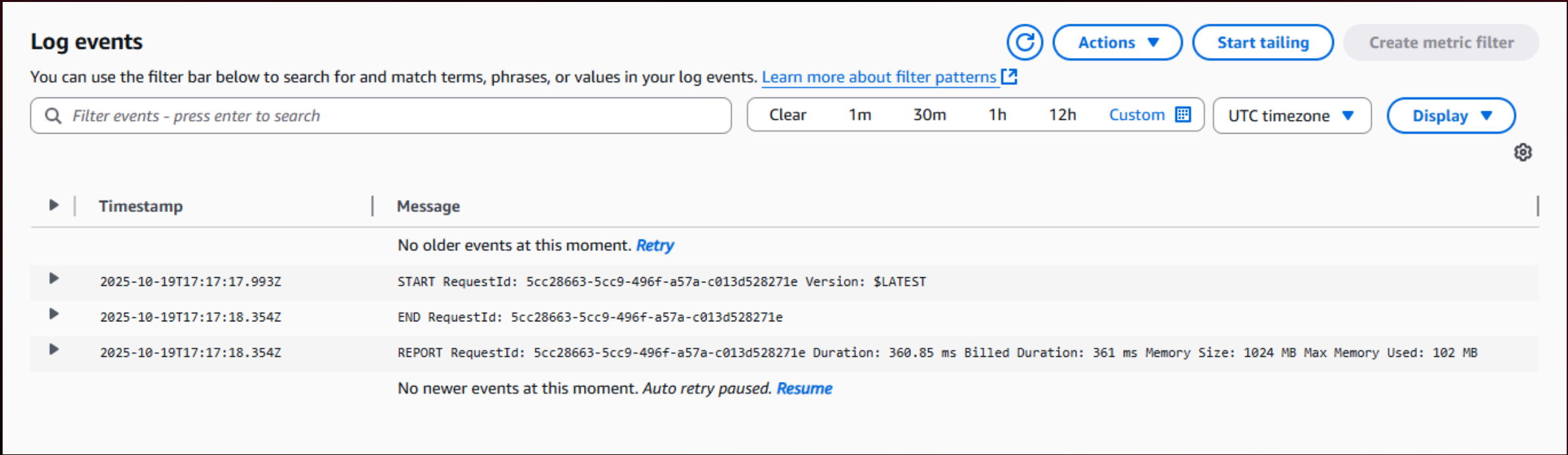
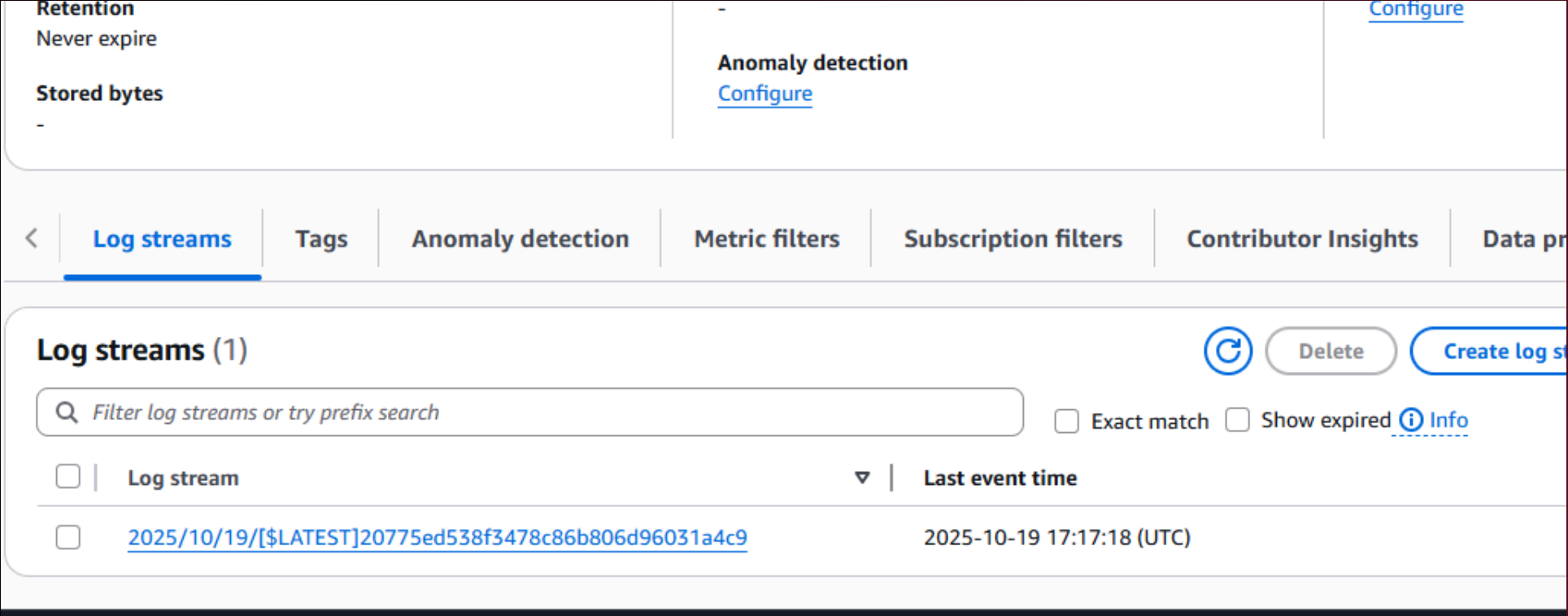
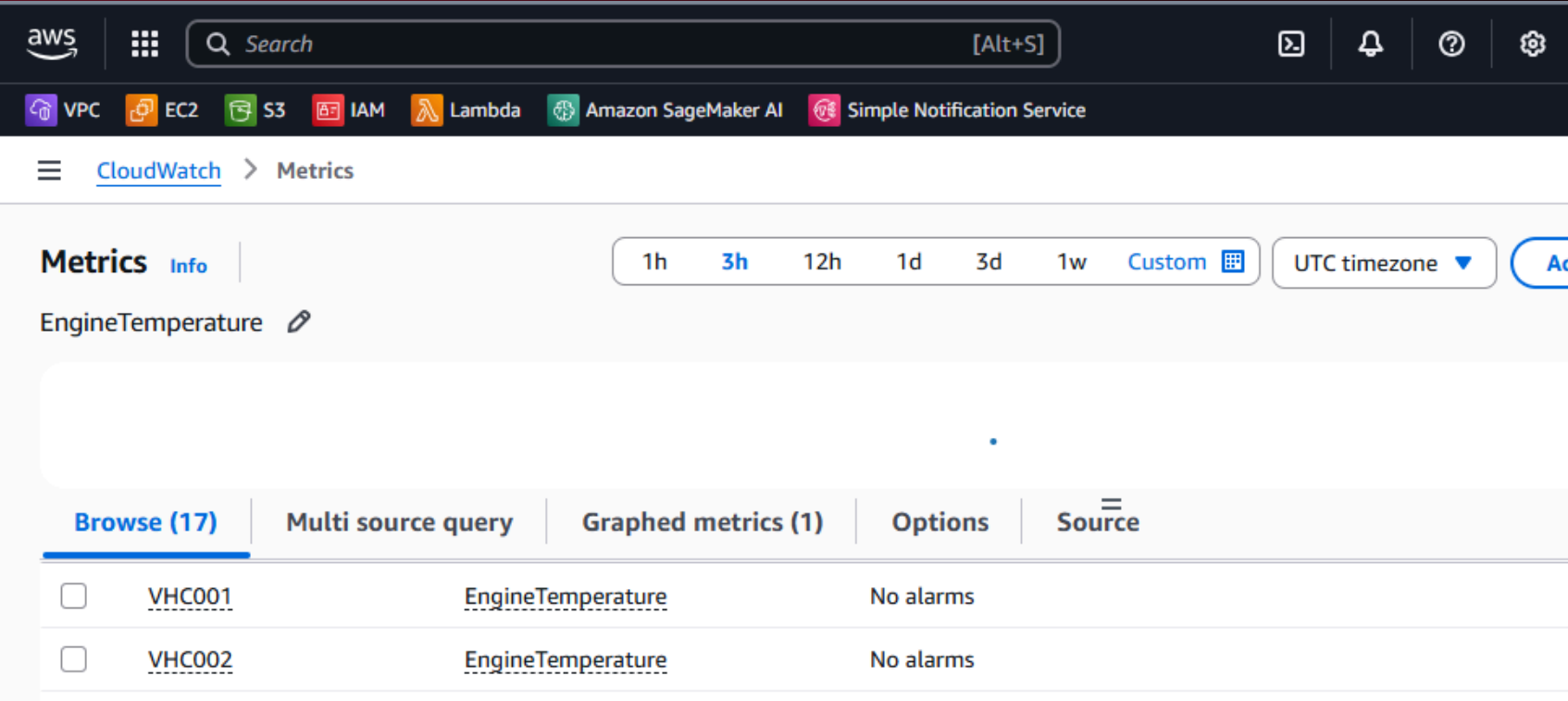
A green status bar indicates the task is 'Completed' with 'Items returned: 2', 'Items scanned: 2', 'Efficiency: 100%', and 'RCUs consumed: 2'.

The table 'fleet-table' shows two items returned. The first item has a 'device_id' of 'VHC002', a 'timestamp' of '19-10-2025 22:47', an 'engine_temp_c' of '124', a 'fuel_level_percent' of '98', and a 'prediction' of '{ "score": { "N": "0.943001449" }, "label": { "S": "POSITIVE" } }'. The second item has a 'device_id' of 'VHC001', a 'timestamp' of '19-10-2025 22:47', an 'engine_temp_c' of '102', a 'fuel_level_percent' of '98', and a 'prediction' of '{ "score": { "N": "0.94451052" }, "label": { "S": "POSITIVE" } }'. The 'prediction' column for both items is highlighted with a red box.

☐	device_id (String)	timestamp (String)	engine_temp_c	fuel_level_percent	prediction
☐	VHC002	19-10-2025 22:47	124	98	{ "score": { "N": "0.943001449" }, "label": { "S": "POSITIVE" } }
☐	VHC001	19-10-2025 22:47	102	98	{ "score": { "N": "0.94451052" }, "label": { "S": "POSITIVE" } }

After uploading raw files in S3, **Dynamo DB** makes it a **clear organised file** and Sagemaker checks on S3 and predicts **"POSITIVE"**.

CLOUD WATCH LOGS



Cloud watch shows the **logs** and **metrics** of the events done recently.

PROJECT SUMMARY

- Designed and implemented an end-to-end serverless **fleet monitoring predictive maintenance** system using AWS.
- Built an **automated workflow** across S3, Lambda, SageMaker, DynamoDB, and SNS.
- Achieved real-time data processing, prediction, alerting, and secure storage.
- Integrated IAM and KMS to ensure **secure access** and **data protection**.
- The system is scalable, cost-effective, and suitable for large fleet operations.
- Full code and detailed documentation are available in [GitHub-Repository](#)

THANK YOU

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